

Homework 1

Lectures 1 through 4 • 100 points

Use the concepts and methods from the first four lectures to describe data, assess study designs, and calculate or simulate probabilities. All scenarios and data in this assignment are fictional.

Submit answers to all 10 questions, your Python and R scripts, and the requested output and figures. Questions 4–6 require hand calculations: show formulas and intermediate steps; a calculator is allowed. For coding questions, explain what the results mean. No hypothesis tests, confidence intervals, or regression models are required.

Use homework1_clinic.csv for Questions 7 and 8. The accompanying README defines every variable. Retain at least four decimals during calculations; report final probabilities to four decimals and numerical summaries to three decimals. Use the specified simulation seeds and report your own output.

Question 1 Population and sample

Conceptual • 8 points

A clinic has 2,000 adults eligible for a sleep program during a specified year. Researchers randomly select 100 of these adults from the complete eligibility list; all participate. After four weeks, 35 report improvement. Everyone receives the same program, and there is no comparison group.

- Identify the target population and the sample. (2 points)
- Define the population response proportion p and the sample statistic $\hat{p}$. Give the observed statistic. (2 points)
- Classify “35% of these 100 adults improved” and “What proportion of all 2,000 eligible adults would improve under this program?” as descriptive or inferential. (2 points)
- Does the observed 35% establish that the program caused improvement? Explain what comparison is missing. (2 points)

Question 2 Study designs and sources of error

Conceptual • 10 points

- Name the design in each scenario: (i) survey current shift work and current symptoms once; (ii) use archived records to identify workers free of the disorder in 2020, classify their 2020 exposure, and ascertain new cases through 2025; (iii) select 80 cases and 160 controls, then review earlier exposures; (iv) randomly assign 120 volunteers to counseling or usual care. (4 points)
- In scenario (iv), distinguish random assignment from random sampling. In scenario (iii), explain why 80/240 is not an estimate of population prevalence. (2 points)
- Distinguish allocation concealment from blinding by stating when each operates and one problem it addresses. (2 points)
- An observational survey measures symptoms differently in the exposed and unexposed groups. Name the type of bias and explain whether increasing the sample size alone fixes it. (2 points)

Question 3 Variables and graphical summaries

Conceptual • 8 points

- a. Classify each variable by its scientific meaning: participant ID stored as 01–12; severity recorded as Mild, Moderate, or Severe; age rounded to whole years; number of prior visits; waiting time in minutes; improvement recorded as 0 or 1. Use nominal identifier, ordinal categorical, quantitative discrete, quantitative continuous, or binary categorical as appropriate. (3 points)
- b. Explain why coding severity as 1, 2, and 3 does not by itself justify treating differences between categories as equal. (1 point)
- c. Choose a suitable graph for each goal: compare numbers in the Yes and No program categories; display the shape of waiting times; show quartiles and unusually long waits; examine paired age and waiting time measurements. (4 points)

Question 4 Center spread and units

Hand calculation • 14 points

The nine observed waiting times, in minutes, are 2, 4, 4, 6, 8, 10, 10, 12, and 25. Treat these as a sample. Use the Lecture 3 linear interpolation convention for quartiles: $h = 1 + (n - 1)p$, with interpolation between the surrounding ordered observations when needed.

- a. Calculate the mean, median, and all modes. (3 points)
- b. Show the deviations from the mean and their squared values. Calculate the sum of squared deviations, sample variance, and sample standard deviation. State their units. (4 points)
- c. Calculate Q1, Q3, IQR, and the two $1.5 \times \text{IQR}$ fences. Identify any flagged value and the actual lower and upper boxplot whiskers. (4 points)
- d. Let $Y = 60X$ convert minutes to seconds. Use the transformation rules to obtain the mean, variance, and SD in seconds. Calculate the sample coefficient of variation and explain whether converting units changes it. (3 points)

Question 5 Joint conditional and independent events

Hand calculation • 12 points

Let A denote working the night shift and B denote having current sleep symptoms. The following is a joint probability table for a hypothetical population.

Shift category	Symptoms B	No symptoms B ^c	Total
Night shift A	0.18	0.12	0.30
Other shift A ^c	0.14	0.56	0.70
Total	0.32	0.68	1.00

- Find $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and $P(A^c)$. Show the addition rule for the union. (4 points)
- Calculate $P(B | A)$ and $P(A | B)$. Explain why the denominators differ. (3 points)
- Determine whether A and B are independent using the multiplication criterion. (2 points)
- Are A and A^c mutually exclusive? Are they independent? Support both answers with probabilities. (3 points)

Question 6 Total probability and Bayes theorem

Hand calculation • 12 points

A hypothetical population is partitioned into three nonoverlapping age groups. Each person belongs to exactly one group. G is a specified new health outcome during the next five years.

Age group	Population proportion	P(G age group)
Under 50	0.50	0.02
50–64	0.30	0.06
65 or older	0.20	0.12

- Use the law of total probability to calculate $P(G)$. Show the weighted terms. (4 points)
- Under this model, calculate the expected number with G among 2,500 people. Explain whether this is a guaranteed observed count. (2 points)
- Use Bayes theorem to calculate the probability that a person is 65 or older given that G occurs. (4 points)
- Explain why the unweighted mean of 0.02, 0.06, and 0.12 is not the population risk for the stated age distribution. (2 points)

Coding data

homework1_clinic.csv has one row per fictional participant. Program participation was chosen at baseline. Waiting time is a baseline measurement; improved records the four-week outcome. Missing waits are coded NA. Preserve all original records and select observed values separately for each analysis.

Question 7 Exploratory data analysis in Python

Python coding • 10 points

- a.** Read the CSV with ID as text. Preserve the severity order Mild < Moderate < Severe. Print the dataset dimensions, duplicate-ID count, missing-wait count, and negative-wait count. (2 points)
- b.** For observed waits, report n, mean, median, sample variance and SD, Q1, Q3, IQR, fences, and flagged values. Use ddof=1 and NumPy quantile method="linear". Compare the overlapping results with your hand calculations in Question 4. (3 points)
- c.** Produce a labeled histogram using edges 0, 5, 10, 15, 20, 25, 30, and print the bin counts. Use bins closed on the left and open on the right, except the final bin includes 30. Also produce a labeled horizontal boxplot with the $1.5 \times \text{IQR}$ rule. (3 points)
- d.** In two or three sentences, describe the distribution, explain why the mean uses the observed-wait count, and state how you would handle the flagged wait. (2 points)

Question 8 An observational comparison in R

R coding • 8 points

- a.** Read the same CSV. Print the program-by-improved count table. For each program group, report total participants, observed improvement outcomes, number improved, improvement proportion, and number of missing waits. Do not exclude an observed improvement outcome because its waiting time is missing. (3 points)
- b.** Calculate the Yes-minus-No difference in improvement proportions. Express it as both a proportion difference and percentage points. (2 points)
- c.** Make a labeled bar chart of the two improvement proportions with a vertical axis from 0 to 1. (1 point)
- d.** Write a defensible interpretation limited to these participants. Explain why the difference is not automatically a causal effect, naming one plausible baseline confounder and how it could affect both program choice and improvement. (2 points)

Question 9 Checking diagnostic probabilities in Python

Python coding • 8 points

In a hypothetical population, disease prevalence is 0.02, sensitivity is 0.90, and specificity is 0.95. Let D mean disease and T mean a positive test. Assume independent simulated people and the stated conditional test probabilities.

- a. Use `np.random.default_rng(813109)` and $B=100000$. First generate B independent uniforms to assign disease; then generate B new independent uniforms to assign positive tests with probability 0.90 for people with disease and 0.05 otherwise. Print the four counts TP, FN, FP, and TN, with clear labels. (3 points)
- b. Calculate empirical sensitivity, specificity, PPV, and NPV from their appropriate subsets. Compute the corresponding theoretical probabilities using the given model and compare theory with simulation. Briefly explain any differences; simulation is an empirical check rather than a mathematical proof. (3 points)
- c. Holding sensitivity and specificity fixed, calculate the theoretical PPV when prevalence rises to 0.20. Explain the direction of the change. (2 points)

Question 10 Sampling without replacement in R

R coding • 10 points

A group contains eight distinct people labeled 1–8; people 1–3 have a specified allergy. In one trial, select three people uniformly without replacement. The event of interest is that all three selected people have the allergy. Reset the full eight-person population before the next trial.

- a. State the exact event probability as a product of conditional probabilities. Also calculate the corresponding probability for three independent draws with replacement. (2 points)
- b. Use $B=100000$ and `set.seed(813110)` to simulate groups with `sample.int(8, 3, replace=FALSE)`. Then use `set.seed(813111)` and `replace=TRUE` for the second model. Report successes, empirical probabilities, and theoretical probabilities for both models. (3 points)
- c. For the simulation without replacement, report running empirical probabilities after 100, 1,000, 10,000, and 100,000 trials. Plot the running probability against the number of trials, with a horizontal line at the theoretical probability. A logarithmic horizontal axis is acceptable. (3 points)
- d. Explain why the two models have different probabilities. Does the law of large numbers guarantee exact agreement after 25 trials, or that each additional trial moves the estimate closer to the truth? (2 points)